

Local Anesthetics

3rd year BDS

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Objectives; Learning to administer local anesthetics effectively, safely and painlessly.

1. What you're giving?

Therefore we will discuss the pharmacology of local anesthetics.

2. Who you're giving it to?

We will discuss how to evaluate patients physically and emotionally. This will involve some physiology. Emergency prevention is emphasized.

3. Where to place it?

The anatomy of respective nerves and adjacent structures must be learned.

4. How to place it there?

Painlessly, in the right amount at the proper rate (slowly). The technique is both an art and a science.

5. How to handle emergencies?

Some may occur in your office or elsewhere. As health professionals we should know what to do, especially if we precipitated the event!

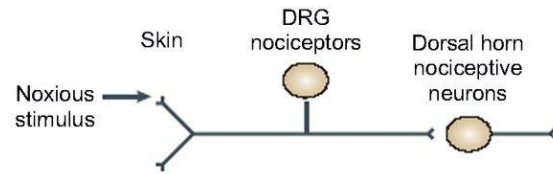
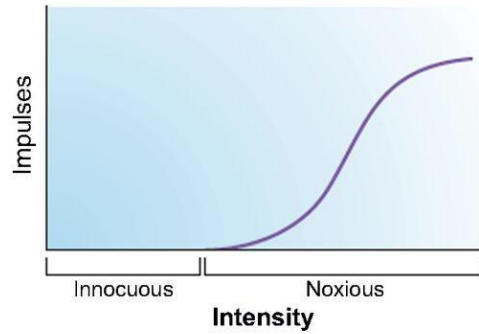
Historical Aspects

- 1845,1853: 1st. Injection of subcutaneous Morphine Solution(London).
- 1903: Addition of Adrenalin to the Cocaine.
- 1904: synthetic Adrenaline(Germany).
- 1905: Synthetic Procaine (Novocain) as Tablets and used with NaCl in Dentistry.
- 1911: 1st. Official Use of LA in Dentistry.
- 1917: using of the LA as Cartridge.
- 1946-48: Synthesis of Lidocaine (Sweden).
- 1976: Development of Articaine (Germany).
- 1981: Intraligamentary Injection.

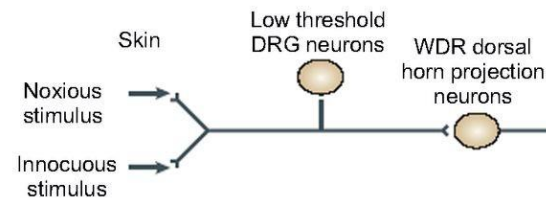
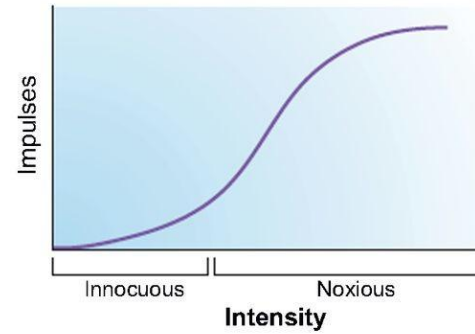
Basis of theories of pain perception:

- Several theoretical works have been proposed to explain concept of pain
- The current definition of pain, established by the International Association for the Study of Pain (IASP) in 1986,
- **Pain** as “an unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of tissue damage, or both.”
- A number of theories have been postulated to describe mechanisms underlying pain perception (the Specificity, Intensity, Pattern, and Gate Control Theories of Pain)

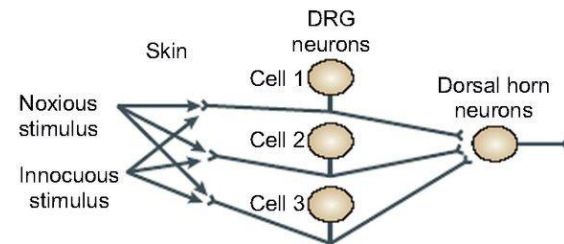
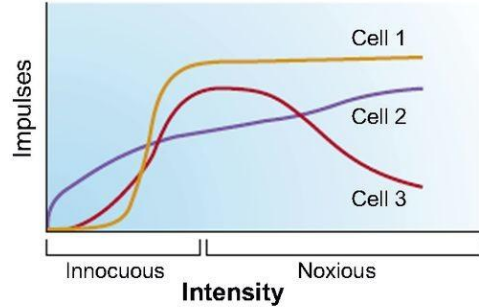
A Specificity theory



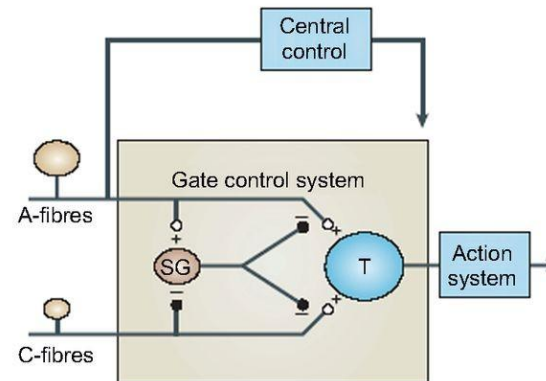
B Intensity theory



C Pattern theory



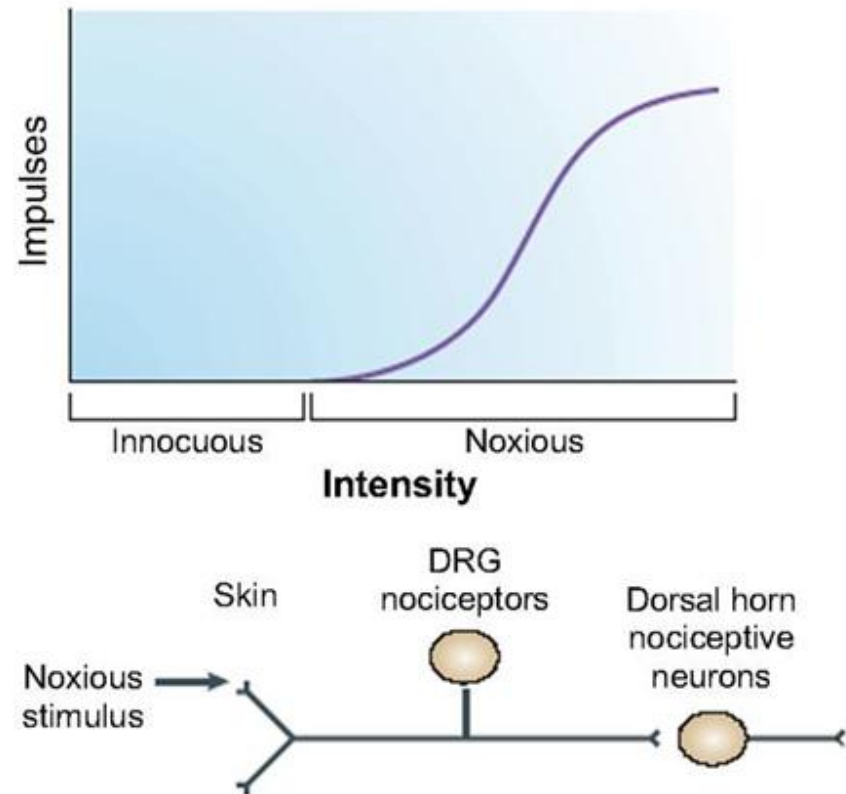
D Gate control theory



SPECIFICITY THEORY OF PAIN

- The presence of dedicated pathways for each somatosensory modality.
- A specific receptor and associated sensory fiber (primary afferent) that is sensitive to one specific stimulus (Dubner et al. 1978).
- each modality (touch and pain) is preset in separate pathways.
- Touch and pain stimuli are encoded by specialized sense organs.
- Impulses for each modality are transmitted along distinct pathways, which project to touch and pain centers in the brain, respectively (DRG, dorsal root ganglion).

A Specificity theory

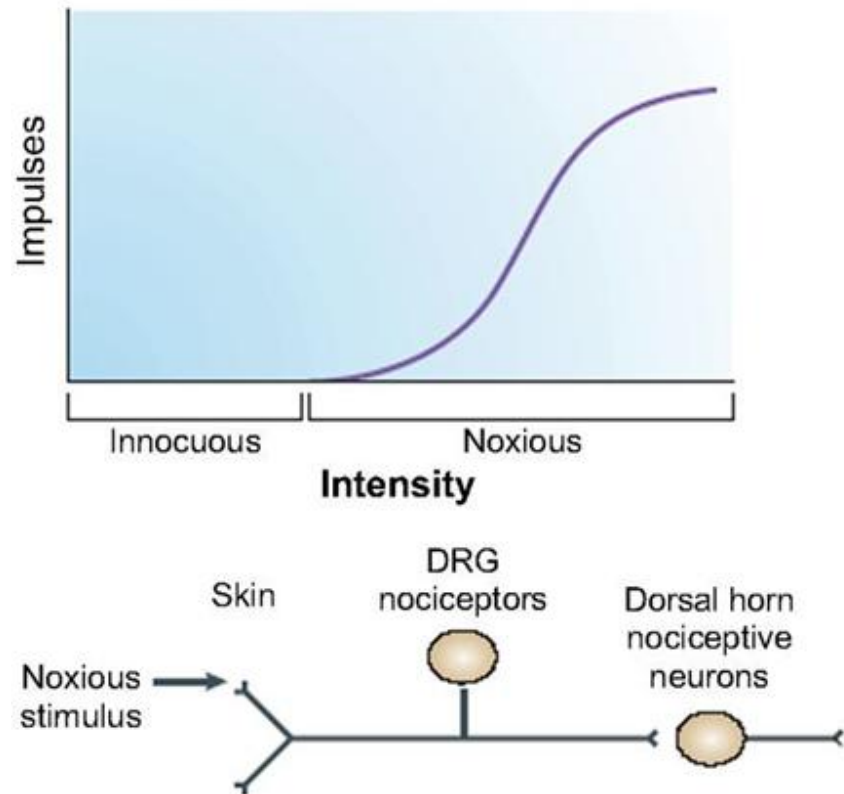


SPECIFICITY THEORY OF PAIN

The **Specificity Theory**

- Pain has its own specific sensory system with receptors, pathways, and centers.
- Stimulus → Pain receptor → Brain = Pain experience.
- Limitation: does not explain psychological influences on pain.

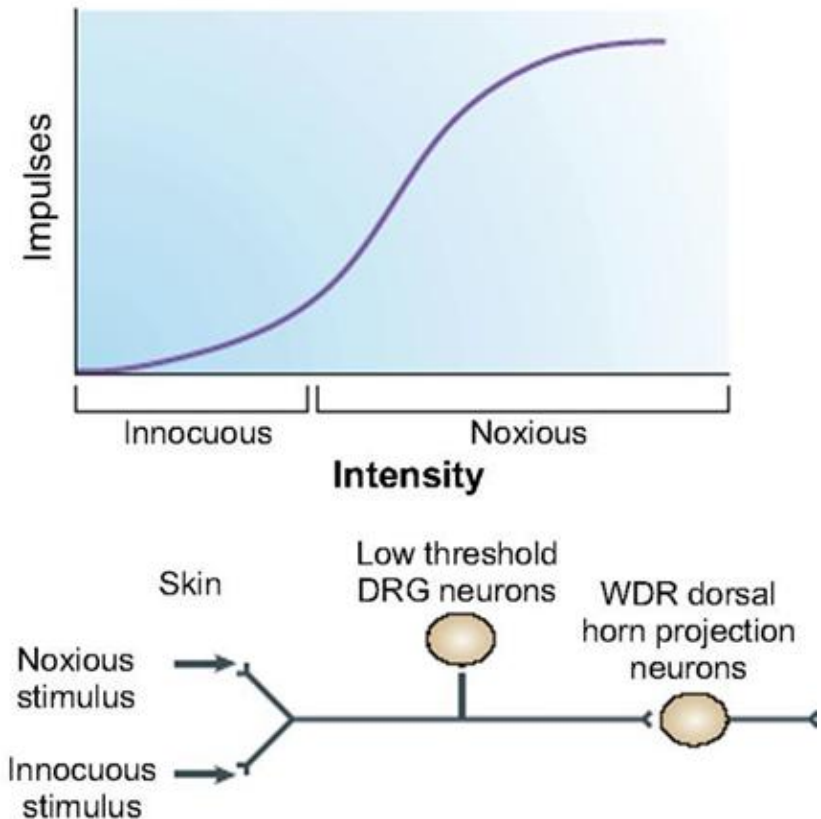
A Specificity theory



INTENSITY THEORY OF PAIN

- Intensive (or Summation) Intensity Theory) has been postulated at several different times throughout history.
- Repeated subthreshold stimulation or suprathreshold hyperintensive stimulation could cause pain.
- there are no distinct pathways for low- and high-threshold stimuli. Rather, the number of impulses in neurons determines the intensity of a stimulus.

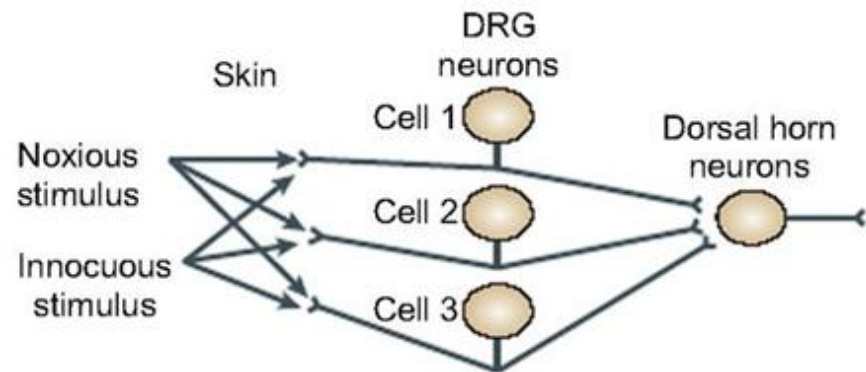
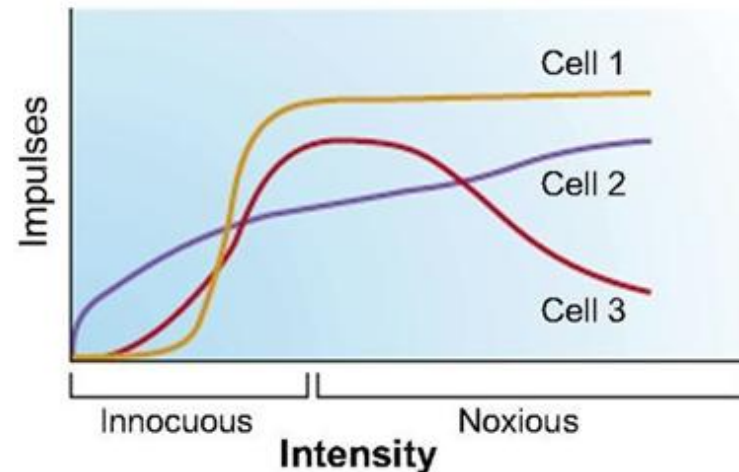
B Intensity theory



PATTERN THEORY OF PAIN

- Pattern Theory of Pain suggests that somatic sense organs respond to a dynamic range of stimulus intensities.
- Different sense organs have different levels of responsivity to stimuli.
- A population code or the pattern of activity of different neurons determine the modality and location of the stimulus

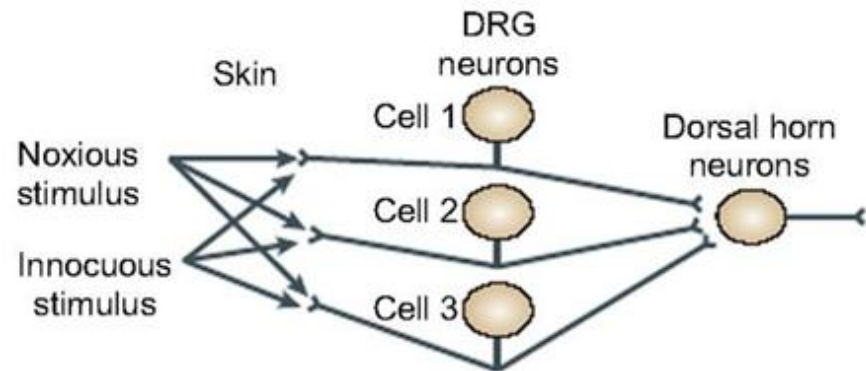
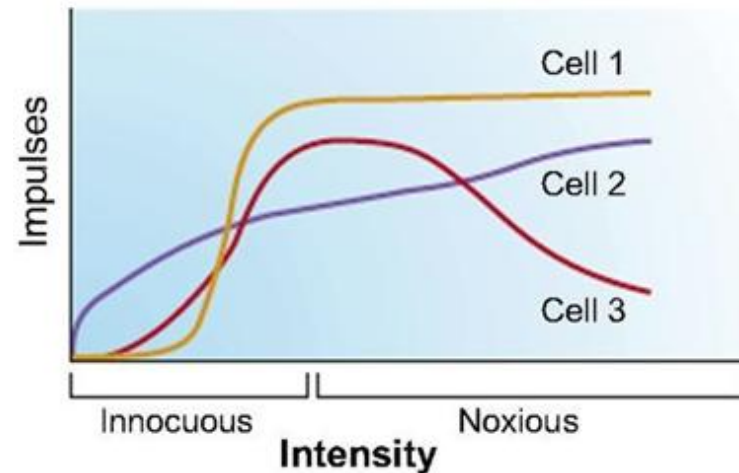
C Pattern theory



PATTERN THEORY OF PAIN

- Pattern Theory results from the intensity and pattern of stimulation of receptors, not from specific "pain receptors."
- "Strong or abnormal stimulation → pain signals."

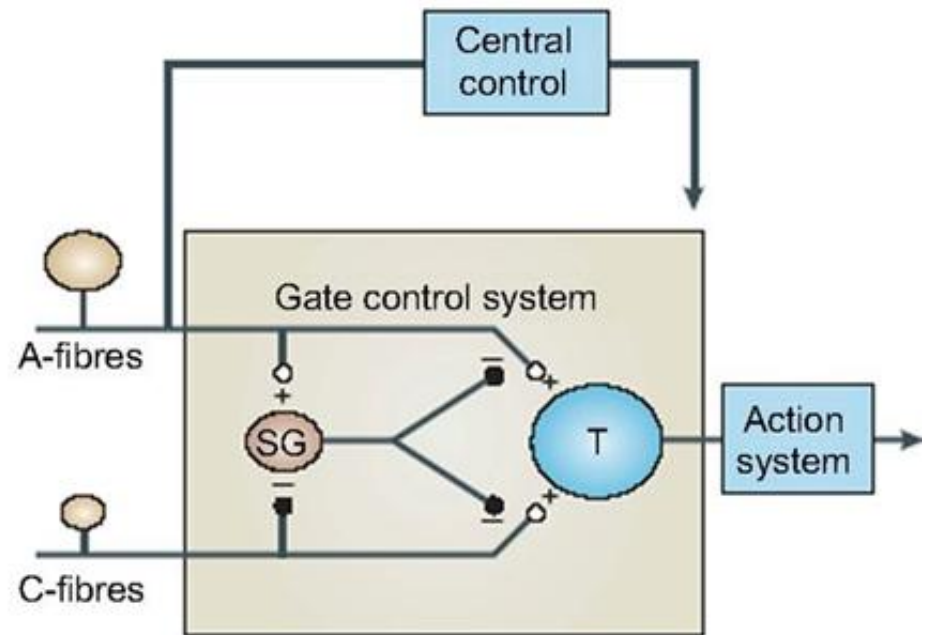
C Pattern theory



GATE CONTROL THEORY OF PAIN

- Theory proposes that both large (A-fibers) and small (C-fibers) synapse onto cells in the substantia gelatinosa (SG) and the 1st central transmission (T) cells.
- The inhibitory effect exerted by SG cells onto the primary afferent fiber terminals at the T cells is increased by activity in A-fibers and decreased by activity in C-fibers.
- The central control trigger is represented by a line running from the A-fiber system to the central control mechanisms; these mechanisms, in turn, project back to the Gate Control system.

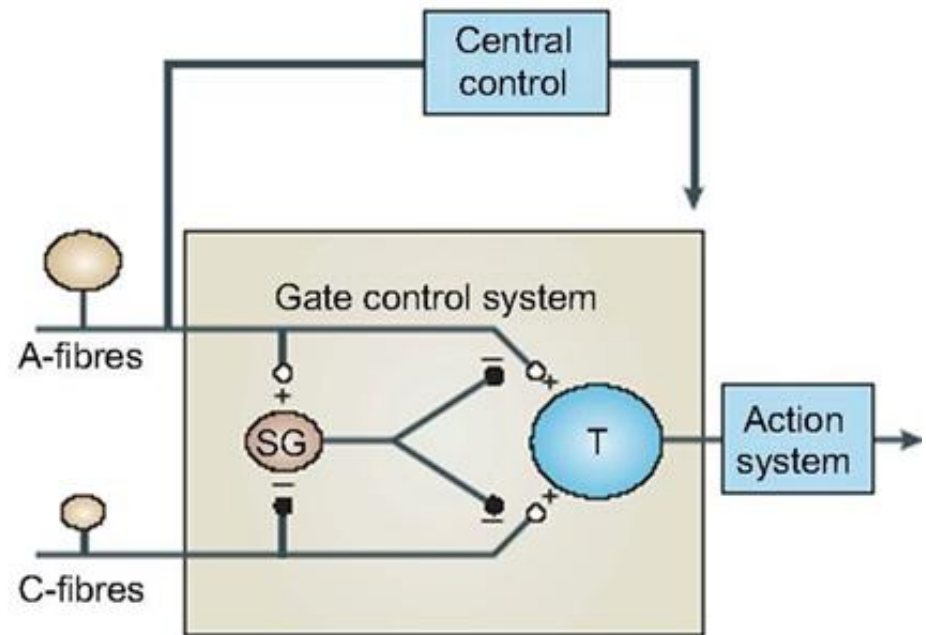
D Gate control theory



GATE CONTROL THEORY OF PAIN

- Gate Control Theory (Melzack & Wall, 1965)
- Pain signals are modulated in the dorsal horn of the spinal cord.
- A “gate” mechanism in the substantia gelatinosa can either block or allow pain impulses to pass to the brain.
- Non-painful stimuli (touch, pressure) can close the gate → reducing pain perception.
- Explains psychological factors: anxiety opens the gate, relaxation closes it.

D Gate control theory



Pain perception versus pain reaction

- **Pain perception;** it is the physioanatomical process, whereby an impulse is generated, subsequent to application of an adequate stimuli, and is transmitted at high centers in brain.
- **Pain reaction;** it is psychophysiological process which represents individual's experience of the unpleasant process of pain perception.
- **Pain threshold;** interpreted as being inversely proportional to pain reaction
- Individual with high pain-threshold considered to be hyporeactive TO noxious stimulus, as opposed to those who have low pain-threshold (hyperactive response).
- **pain-threshold affected** by emotions, fatigue, stress, age, gender, fear, apprehension, anxiety.

Mechanism of Pain Transmission in Dentistry

- **Stimulation** of nerve endings (A-delta and C fibers).
- **Transmission** through peripheral nerves → trigeminal nerve → CNS.
- **Perception** in higher brain centers (thalamus, cortex).
- **Reaction:** emotional and behavioral responses.

Pain Control Theories Applied in Dentistry

- Psychological aspect (Gate Control Theory): Anxiety management, distraction, and reassurance reduce pain.
- Physiological aspect (Specificity/Pattern): Adequate LA blocks pain receptors and transmission.
- Clinical integration: Combination of pharmacological (LA, sedation) and non-pharmacological methods (communication, relaxation, behavioral techniques).

Methods of local anesthesia:

- Mechanical trauma (compression of tissues)
- Low temperature
- Anoxia
- Chemical irritants
- Neurolytic agents such as alcohol and phenol
- Chemical agents such as local anesthetics

Definitions:

- **Analgesia:** Pain killing.
- **Anaesthesia:** it is a loss of all forms sensation including touch, temperature, pressure and may be accompanied by impairment of motor function.
- **Classification:** 1- local (LA)limited area. 2- general (GA).
- Thus if a local anaesthetic drug is injected in the nerve, all perception of pain is lost but other sensation, such as pressure or vibration are still felt.
- **Concept of local anaesthesia:** Local anaesthesia is obtained by injecting an anaesthetic drug near the sensory nerves in limited area to prevent the conduction of pain impulses to the brain without loss of consciousness, it is reversible.
- **General anesthesia:** a state of unconsciousness.

DENTAL USES OF LOCAL ANAESTHESIA:

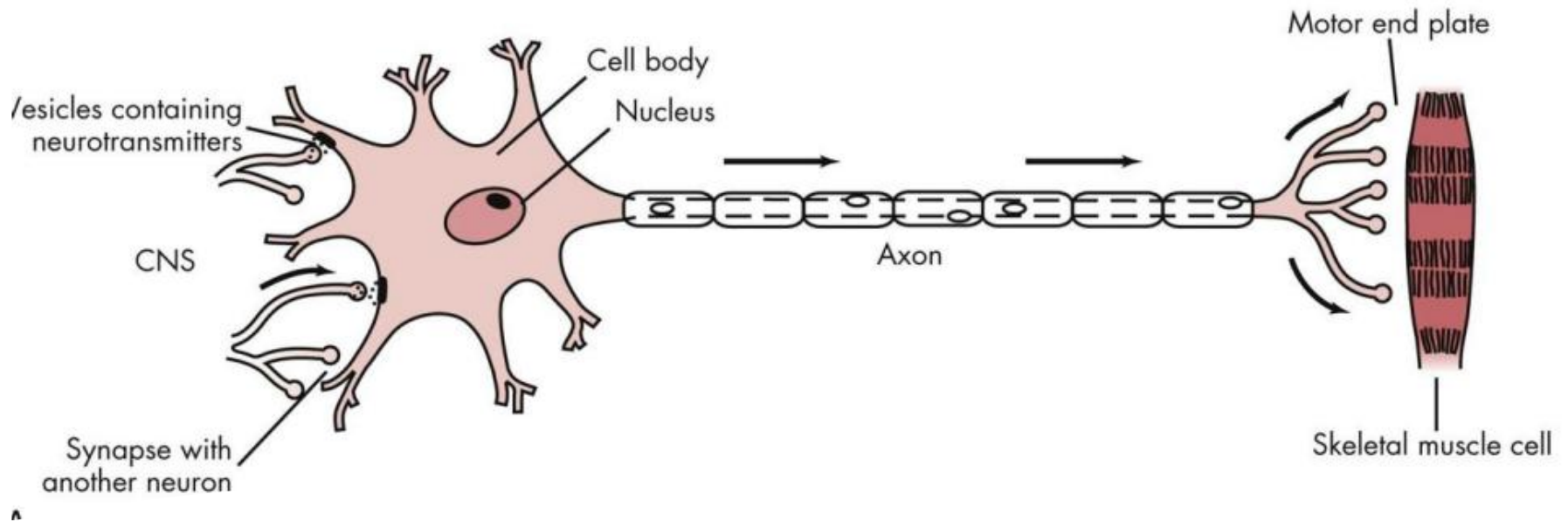
- Elimination of pain during treatment.
- Diagnostic purpose- L.A. can be very helpful with some diagnostic problems because if an injection is given to block nerve conduction in a particular region and the pain is relieved , then it can be deduced that the causative lesion is in the tissue innervated by that nerve.
- To reduce haemorrhage- this is due to the action of vasoconstrictor .
- Healing Injections.....etc.

Advantages of LA over GA:

- 1) Safety- L.A. safer than G.A.
- 2) Ease of administration.
- 3) Cooperation of the patient - e.g. the patient can be asked to occlude his teeth during treatment to check for any high point.
- 4) Reduced bleeding during surgical treatment.
- 5) Economical.
- 6) The equipment required is limited.
- 7) Easily transportable.
- 8) The patency of airway is not impaired.
- 9) When the patient is unfit for G.A.

Contraindications of L.A

- 1) Pt. refused L.A because of fear and apprehension.
- 2) Presence of infection in the pathway of the needle.
- 3) If the pt is allergic.
- 4) Uncooperative pt due to mental disturbances or epilepsy or other conditions.
- 5) In case of anatomical anomalies which make injection of L.A difficult or impossible.
- 6) If major surgery is to be performed.



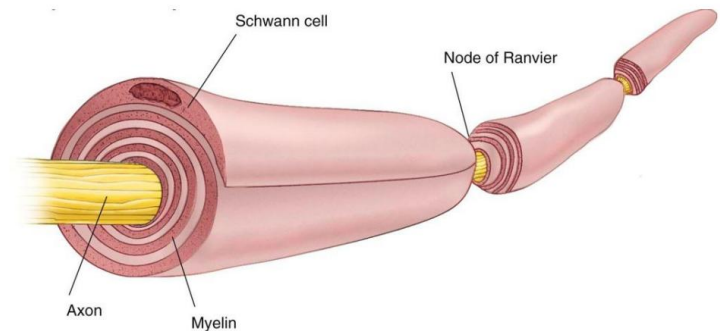
Nerve anatomy and physiology

The neuron, or nerve cell, is the structural unit of the nervous system. It is able to transmit messages between the central nervous system (CNS) and all parts of the body. There are two basic types of neuron: sensory (afferent) and motor (efferent).

Biological arrangement of Axons neuron membrane.

- The axon is a thin cable-like structure (100 to 200 nm).
- Most axons are myelinated and surrounded by a concentric myelin sheath layer, produced by a Schwann cell (Kerns, 2008).
- The thickness of the myelin sheath is directly related to the axon diameter.

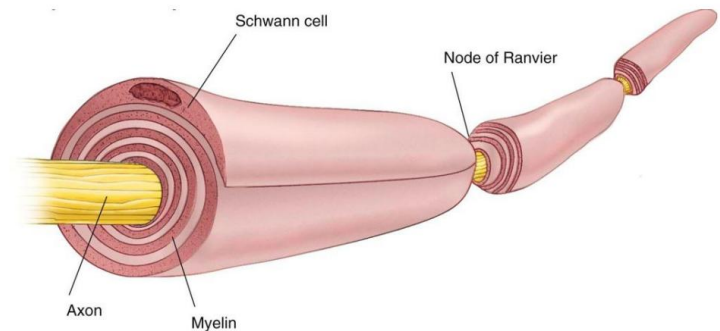
Figure 1-5 Structure of a myelinated nerve fiber.



Biological arrangement of Axons neuron membrane.

- The sheath layers are lipid rich plasma membranes and confer increased rates of transmission and conduction of nerve impulses.
- The span of axonal membrane that is covered by each Schwann cell is called an internode (Topp and Boyd, 2012).
- The space between the Schwann cells is known as node of Ranvier, where ion channels are present, and are responsible for continuous propagation of electrical impulses at the subsequent node of Ranvier (Kerns, 2008; Topp and Boyd, 2012)

Figure 1-5 Structure of a myelinated nerve fiber.



Classification of Peripheral Nerves According to Fiber Size and Physiologic Properties

Fiber Class	Subclass	Myelin	Diameter, μ	Conduction Velocity, m/s	Location	Function
A	alpha	+	6-22	30-120	Afferent to and efferent from muscles and joints	Motor, proprioception
	beta	+	6-22	30-120	Afferent to and efferent from muscles and joints	Motor, proprioception
	gamma	+	3-6	15-35	Efferent to muscle spindles	Muscle tone
	delta	+	1-4	5-25	Afferent sensory nerves	Pain, temperature, touch
B		+	<3	3-15	Preganglionic sympathetic	Various autonomic functions
C	sC	–	0.3-1.3	0.7-1.3	Postganglionic sympathetic	Various autonomic functions
	d gammaC	–	0.4-1.2	0.1-2.0	Afferent sensory nerves	Various autonomic functions; pain, temperature, touch
From Berde CB, Strichartz GR: Local anesthetics. In Miller RD, editor: Anesthesia, ed 5, Philadelphia, 2000, Churchill Livingstone, pp 491–521.						

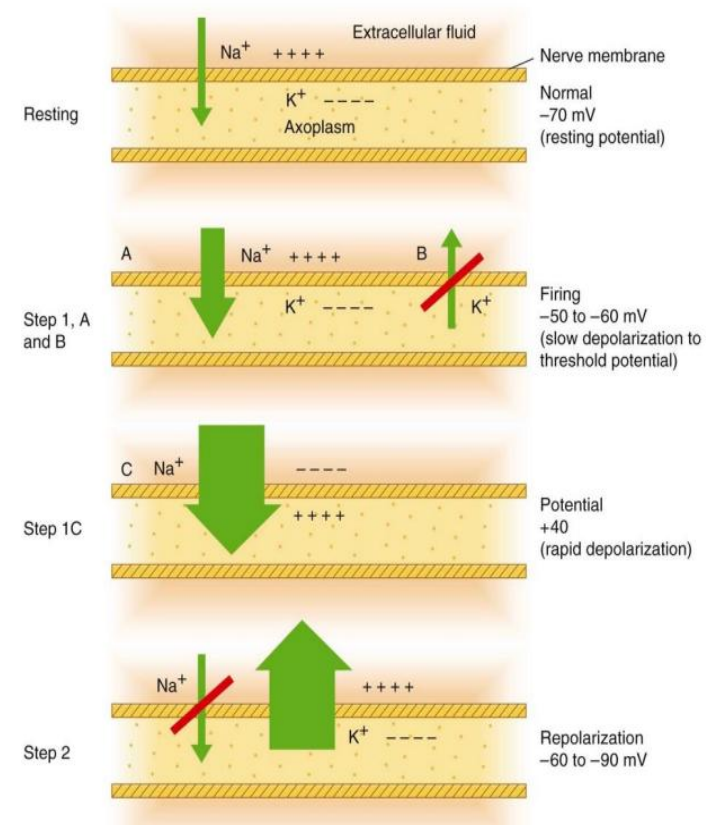
Heavily myelinated fibres are associated with the transmission of motor and proprioceptive impulses whereas less myelinated and unmyelinated fibres convey sensory signals.

Physiology of Action potential

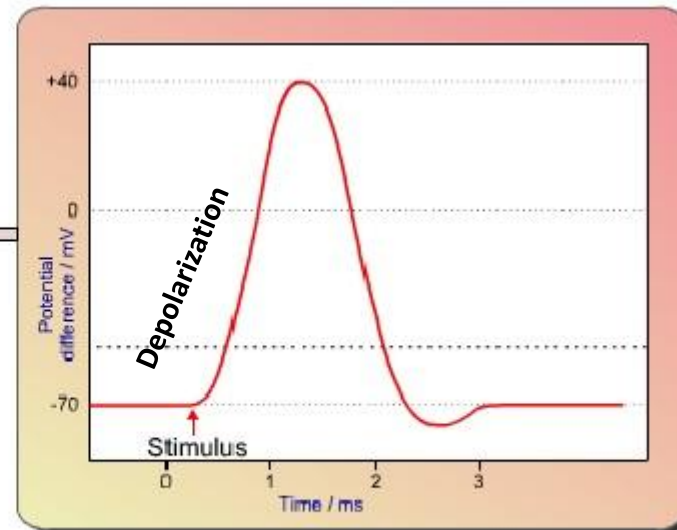
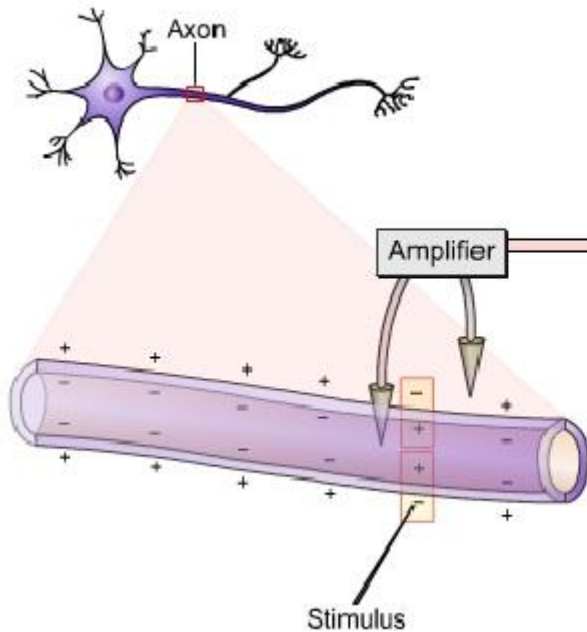
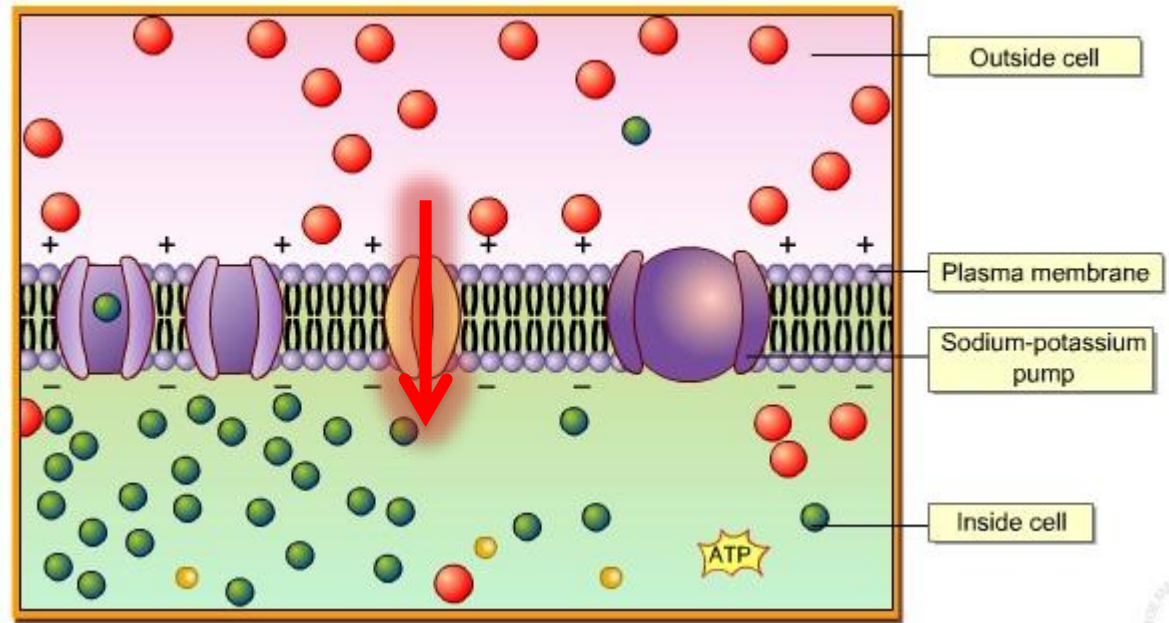
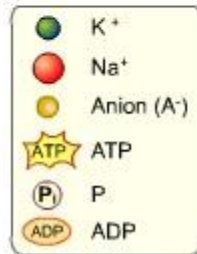
- In the peripheral nervous system, the neurones create, amplify and propagate electrical impulses along their axon terminals These impulses are called action potentials (Topp and Boyd, 2012).

Action potential & nerve conduction.

- An action potential occurs in response to alterations in potassium and sodium ion gradients across the cell membrane in response to tissue damage or mechanical stimulation in the nociceptive fibres and proprioceptive fibres, respectively.
- At rest the neurone's cell body maintains its intracellular membrane potential at -70 mV by retaining higher concentration of potassium (K^+) ions as opposed to low concentration of sodium ions (Na^+)
- During excitation, sodium channels open and allow an influx of Na^+ ions to produce an action potential by raising the membrane potential to a threshold potential of -50 mV causing membrane depolarization.
- Afterwards, previously opened sodium channels become inactive and potassium channels open to repolarize the membrane to its resting potential state and generate a new action potential at another site



Action potential



Resting State

- In its resting state, the nerve membrane is;
 - Slightly permeable to sodium ions (Na^+)
 - Freely permeable to potassium ions (K^+)
 - Freely permeable to chloride ions (Cl^-)

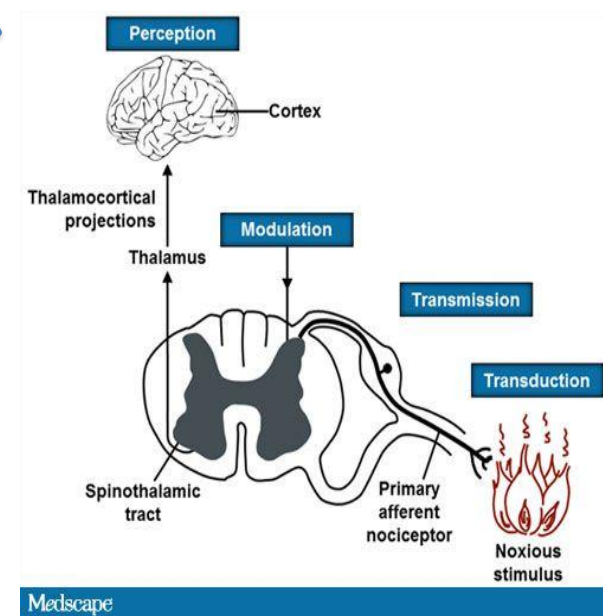
Is the condition that results when sensory transmission (action potential) from a local area of the body to the CNS is blockade

Block nerve conduction of sensory impulse

CAUSED

Local Anesthetic
(LA)

Local Anesthesia



Is a drug that causes reversible local anesthesia and a loss of nociception.

Mode and Site of Action of Local Anesthetics

- LA interfere with the excitation process in a nerve membrane in one or more of the following ways:
 1. Altering the basic resting potential of the nerve membrane
 2. Altering the threshold potential (firing level)
 3. Decreasing the rate of depolarization
 4. Prolonging the rate of repolarization

LA & orofacial pain



- International Association for the Study of Pain (IASP) definition of pain as "An unpleasant sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage"
- When a nerve is stimulated, an impulse is propagated that will be interpreted as pain when it reaches the brain
- Local anesthesia; defined as localized loss of sensation caused by depression of excitation in nerve endings or inhibition of the conduction process in peripheral nerves.

Theories of mechanism of action of local anesthetics

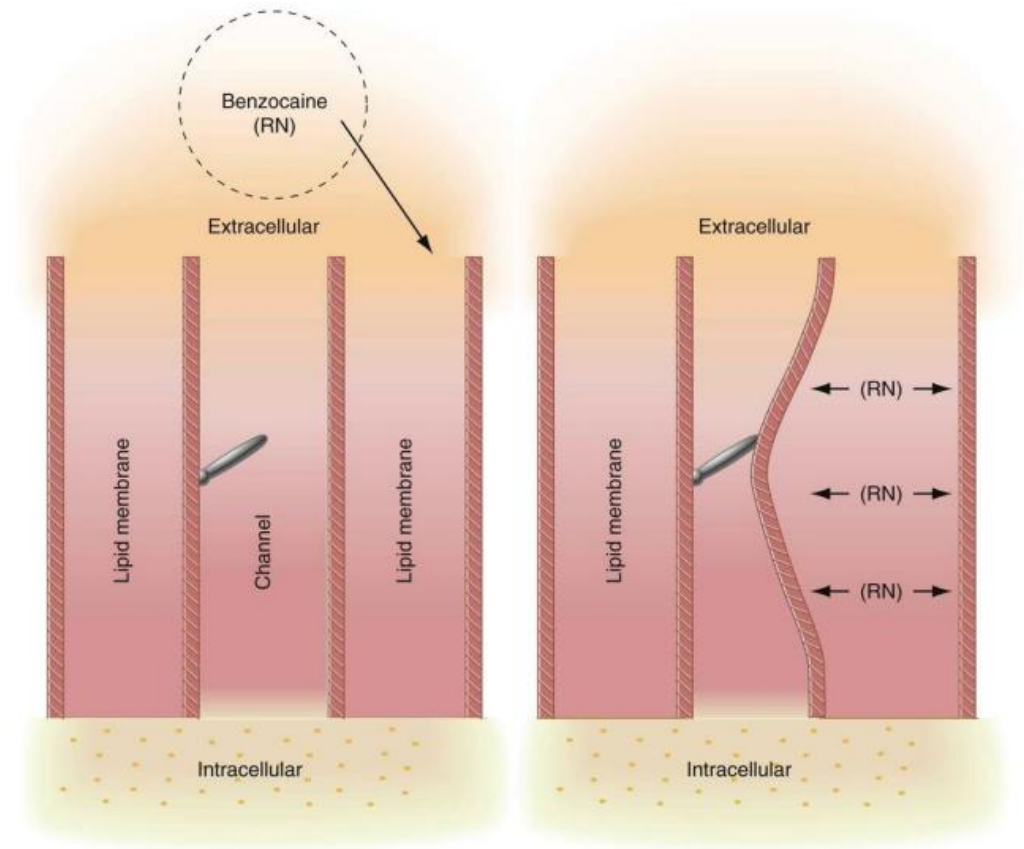


MEMBRANE EXPANSION
THEORY



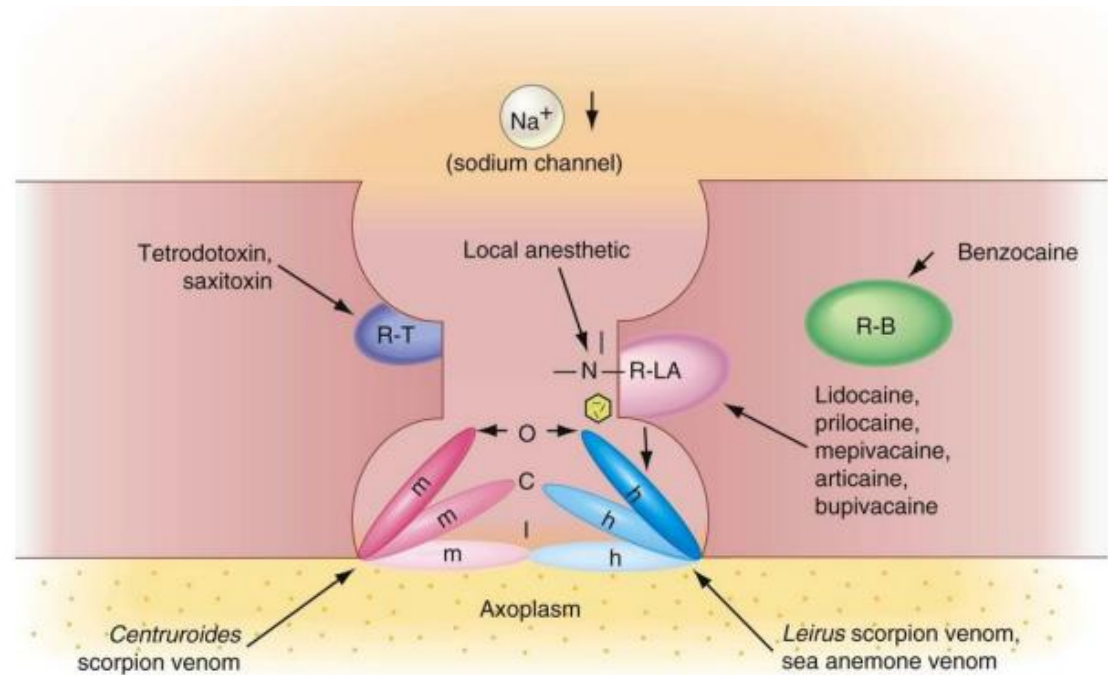
SPECIFIC RECEPTOR
THEORY

Membrane expansion theory



The theory states that LA molecules diffuse to hydrophobic regions of excitable membranes, producing a disturbance of membrane structure, & expanding some critical region(s) in the membrane, and preventing an increase in permeability to sodium ions which leads to inhibition of both sodium conductance and neural excitation.

Specific receptor theory



The theory states that LA act by binding to specific receptors on the sodium channel, however, action not mediated by properties of the cell membrane which prevent nerve excitation by elimination sodium ions permeability.

(A specific receptor site for LA drugs exists in the sodium channel either on its external surface or on the internal axoplasmic surface).

Theories of Mechanism of Action of Local Anaesthetic

- **Membrane Expansion Theory**

- Local anaesthetic molecules are lipophilic.
- They incorporate into the lipid bilayer of the nerve membrane.
- This causes physical expansion of the membrane, leading to narrowing or distortion of sodium channels.
- Result: decreased sodium permeability → no depolarization → conduction block.

- **Specific Receptor Theory (accepted theory)**

- Local anaesthetic molecules bind directly to specific receptor sites within voltage-gated sodium channels.
- Binding occurs on the intracellular side of the sodium channel.
- Blocks sodium influx → prevents depolarization → no action potential.
- Explains the frequency-dependent block (more active nerves are blocked faster).

widely accepted
explanation.

How Local Anaesthetic Work

- Local anaesthetic block pain by interrupting nerve conduction.
- **Mechanism:**
- LA diffuses through the nerve sheath.
- LA binds to sodium channels inside the nerve membrane.
- Prevents sodium influx → no depolarization.
- Conduction block: action potentials cannot propagate.
- Pain signals never reach the brain.
- Order of blockade: Pain → Temperature → Touch → Pressure → Motor (due to different fiber sizes)

Proposed mechanism of action of LA drugs

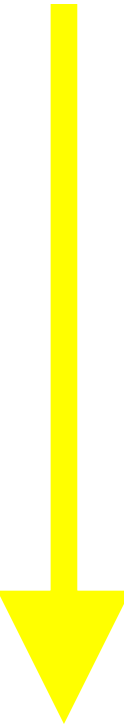
Displacement of calcium ions from the sodium channel receptor site, which permits

Binding of the local anesthetic molecule to this receptor site, which produces

Blockade of the sodium channel and decrease in sodium conductance, which leads to ...

Depression of the rate of electrical depolarization, Failure to achieve the threshold potential level, along with

Lack of development of propagated action potentials (Conduction blockade)



Sequence of clinical anesthesia

Sympathetic block (vasodilatation)



Loss of pain and temperature sensation



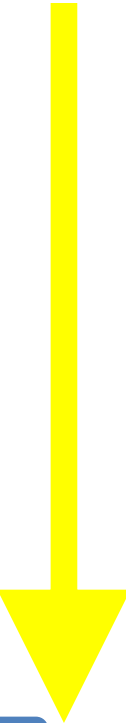
Loss of proprioception



Loss of touch and pressure sensation

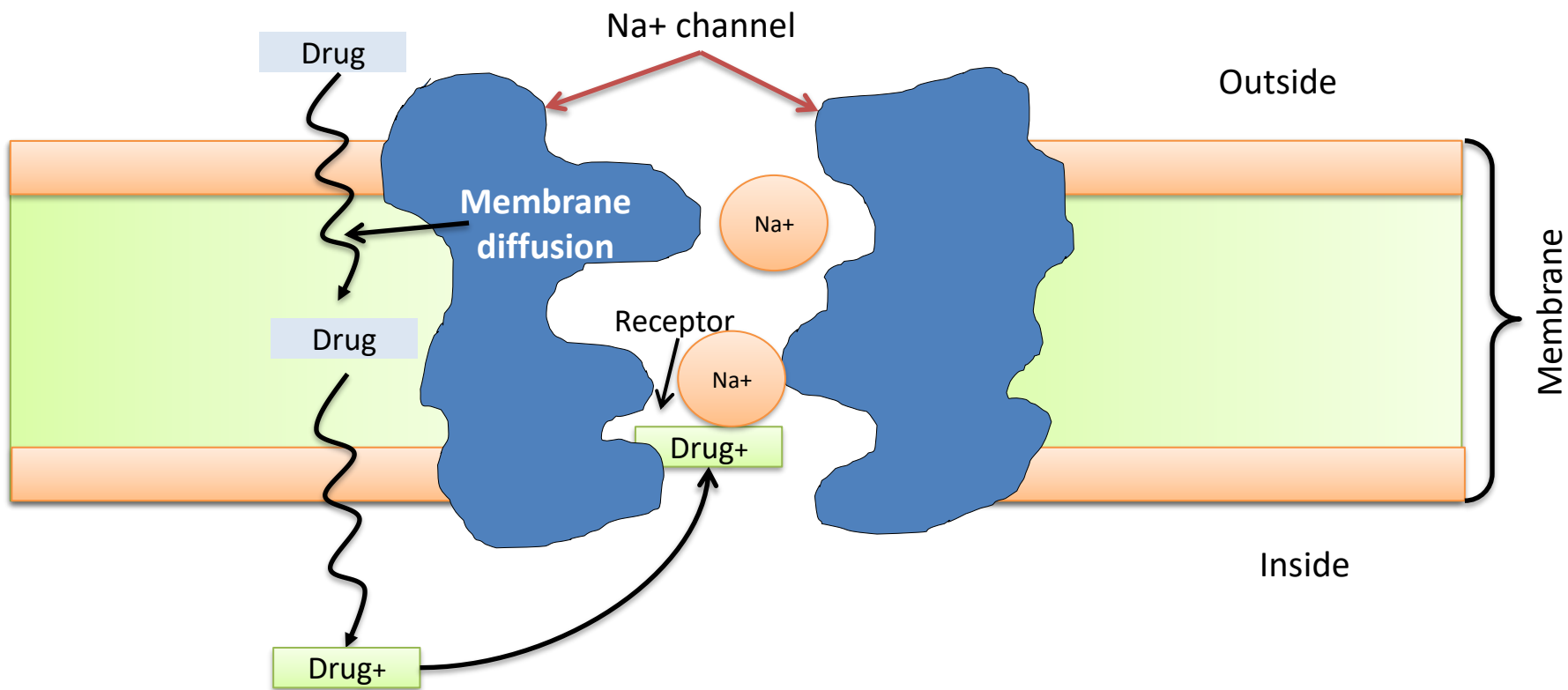


Loss of motor function



Mechanism of action of LA

LAs **block** voltage-dependent sodium channels (at axon) and reduce the influx of sodium ions, thereby preventing depolarization of the membrane and blocking conduction of the action potential.



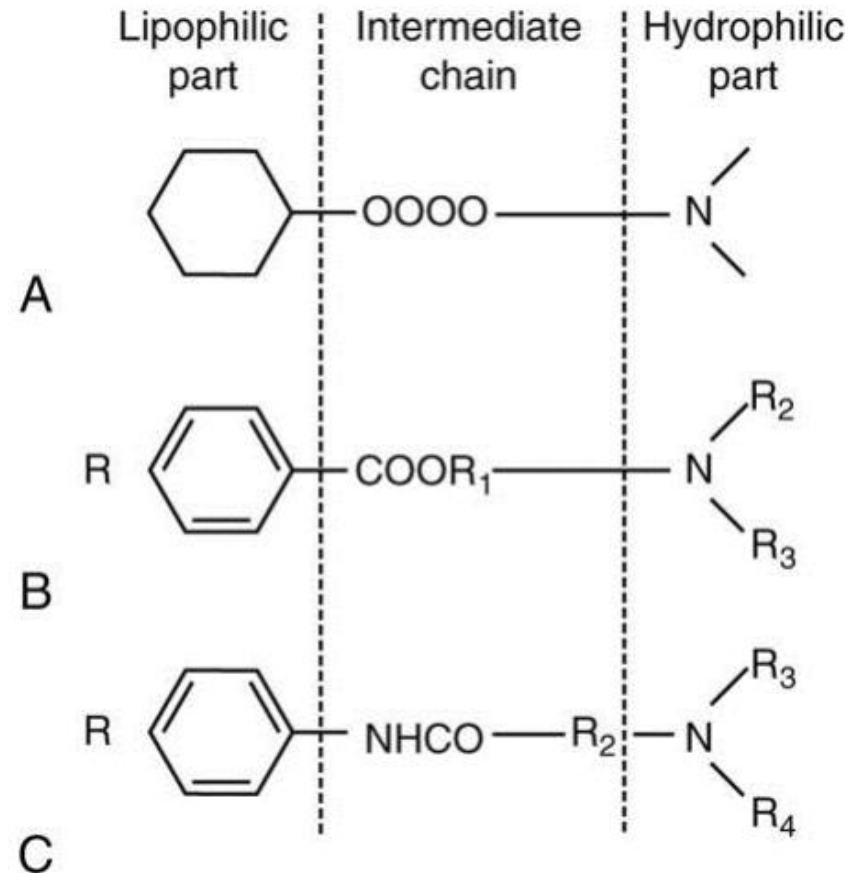
Classification According to Biological Site and Mode of Action

Classification Definition		Chemical Substance
Class A	Agents acting at receptor site on external surface of nerve membrane	Biotoxins (e.g., tetrodotoxin, saxitoxin)
Class B	Agents acting at receptor site on internal surface of nerve membrane	Quaternary ammonium analogs of lidocaine Scorpion venom
Class C	Agents acting by a receptor-independent physico-chemical mechanism	Benzocaine
Class D	Agents acting by combination of receptor and receptor-independent mechanisms	Most clinically useful local anesthetic agents (e.g., articaine, lidocaine, mepivacaine, prilocaine)
Modified from Covino BG, Vassallo HG: Local anesthetics: mechanisms of action and clinical use, New York, 1976, Grune & Stratton. Used by permission.		

Pharmacology of Local Anesthetics

- Most injectable local anesthetics active forms are **tertiary amines**. Only a few (e.g., prilocaine, hexylcaine) are secondary amines
- All local anesthetics are **amphipathic**
- Aromatic in structure, it is derived from **benzoic acid**, aniline, or thiophene
- The hydrophilic part is an amino derivative of ethyl alcohol or acetic acid

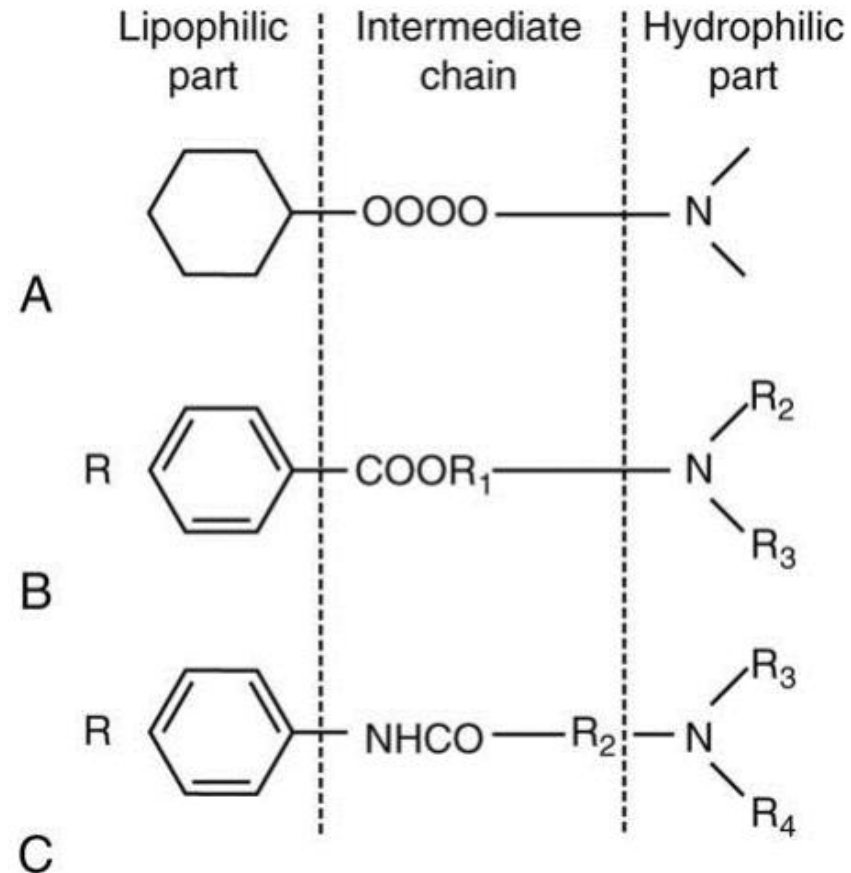
A, Typical local anesthetic. **B**, Ester type. **C**, Amide type.



Pharmacology of Local Anesthetics

- The anesthetic structure is completed by an intermediate hydrocarbon chain containing an ester or an amide linkage

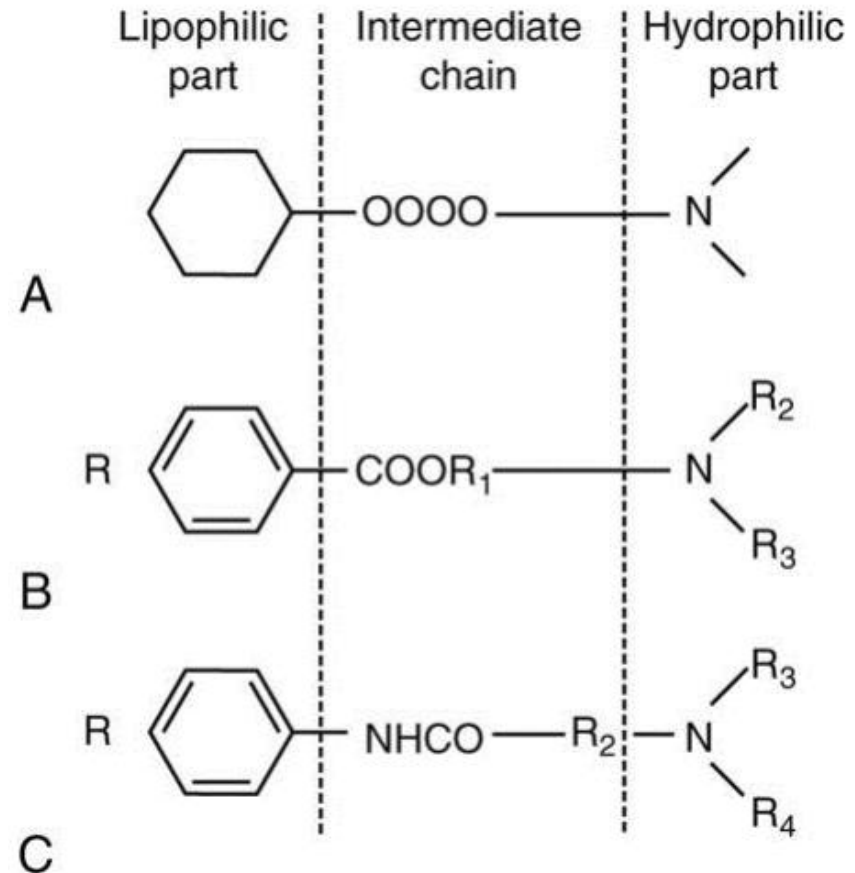
A, Typical local anesthetic. **B**, Ester type. **C**, Amide type.



Pharmacology of Local Anesthetics

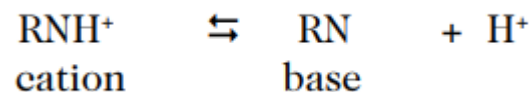
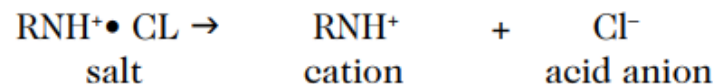
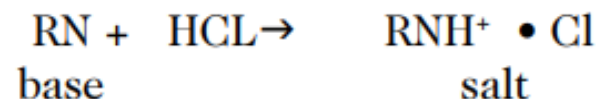
- The nature of the linkage determine mode of biotransformation.
- Ester-linked local anesthetics (e.g., procaine) are readily hydrolyzed in aqueous solution.
- Amide-linked local anesthetics (e.g., lidocaine) are relatively resistant to hydrolysis.
- A greater percentage of an amide-linked drug than of an ester-linked drug is excreted unchanged in the urine

A, Typical local anesthetic. **B**, Ester type. **C**, Amide type.



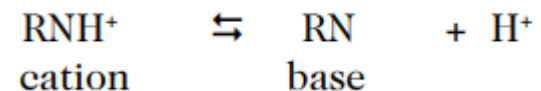
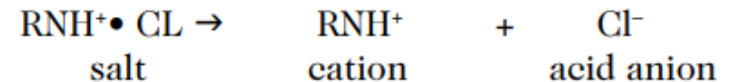
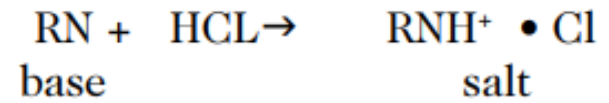
Chemical Properties of LA

- **weak** basic **tertiary amines** (acceptors of hydrogen ion)
- poorly soluble in water
- Unstable when exposed to air.
- **weak base** binds with acids to form a salt



Chemical Properties of LA

- In sterile water or saline, the **local anesthetic salt ionizes to form a quaternary amine cation and an acid anion.**
- The cation [ionized] then, is in an dissociation equilibrium with the base[non-ionized].
- The **proportion** depends **upon the pH** (H⁺ concentration) of the solution and the surrounding tissues as well as **the pKa** (dissociation constant) of **each specific local anesthetic.**



Ionization of Local Anaesthetics

- **Salt Form in Solution**

- Local anaesthetic are usually prepared as **water-soluble hydrochloride salts** (e.g., lidocaine HCl).
- In **sterile water or saline**, they ionize into:
 - **Quaternary amine cation (ionized form)**
 - **Acid anion**

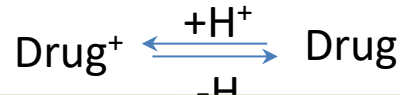
- **Equilibrium Between Ionized and Unionized Forms**

- The **cation (ionized form)** $\leftrightarrow$ **base (non-ionized form)** exist in equilibrium.
- Governed by the **pKa of the anaesthetic** and the **pH of the tissue** (Henderson-Hasselbalch relationship).

Ionization of Local Anaesthetics

- **Role of Each Form**
 - **Non-ionized (base):**
 - Lipid soluble.
 - Diffuses across the nerve membrane.
 - **Ionized (cation):**
 - Water soluble.
 - Active form that binds to the **sodium channel receptor** inside the nerve.
- **Clinical Implication**
 - In normal tissue (pH ~7.4):
 - A balance exists between ionized and unionized forms → effective anaesthesia.
 - In **inflamed or infected tissue (acidic pH ~5–6):**
 - Equilibrium shifts → more ionized form, less non-ionized base.
 - Drug cannot penetrate nerve membrane → **poor anaesthetic effect**.
-  **Points:**
- LA must first **cross the membrane (unionized base)**, then **block sodium channels (ionized cation)**.
- Tissue pH and drug pKa determine the balance → directly affecting **onset and effectiveness**.

The commonly used LA are **weak base** with at least 1 ionizable amine function that can become charged through the gain of a proton.



Thus, both the nonionized and the ionized forms of the drug play important roles, the first in reaching the receptor site and the second in causing the effect.

the ionize (charged) form

The Nonionize (uncharged) form



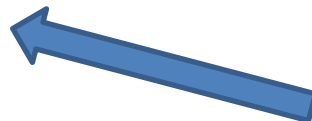
Once the drug has penetrated the lipid barrier and reach its site of action (receptor) it ionized and the ionized form is responsible for LA activity

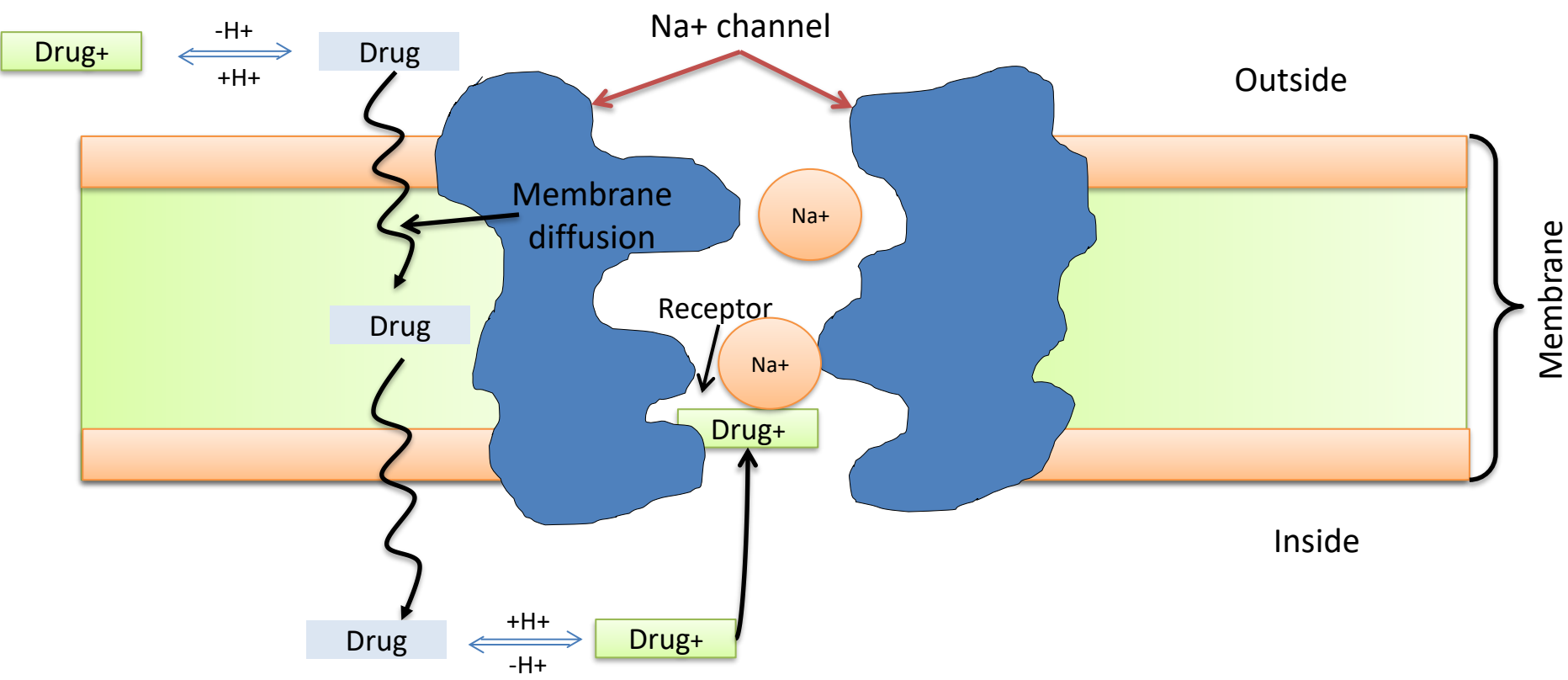


More lipid soluble



It will facilitate its penetration through the cell membrane (bilayer phospholipid)





↑ PH → ↑ Uncharged forms

↓ PH → ↑ Charged forms
Acidosis



Acidosis such as caused by inflammation at a wound partly reduces the action of LAs. This is partly because most of the anesthetic is ionized and therefore unable to cross the cell membrane.

Learning Points

- Local anaesthetic = weak bases → made into salts for solubility.
- They exist in equilibrium between ionized (RNH^+) and unionized (RN).
- Unionized form = crosses nerve membrane.
- Ionized form = blocks sodium channels.
- Effectiveness depends on pKa of drug and pH of tissue.

Pharmacokinetics properties

- Onset, potency and duration of action is dependent on the pKa, lipid solubility, Vasoactivity and their protein binding capacity.

Glossary and terms related to ionisation of LA

Term	Full Word / Meaning	Simple Definition
RNH⁺	Ionised form of local anaesthetic	The charged (acidic) form that cannot cross the nerve membrane but blocks the nerve once inside.
RN	Non-ionised form of local anaesthetic	The uncharged (basic) form that can cross the nerve membrane but must convert to RNH ⁺ inside the nerve to act.
pKa	Acid dissociation constant	The pH at which 50% of the drug is ionised (RNH ⁺) and 50% non-ionised (RN) . Determines how fast the anaesthetic works.
pH	Power of Hydrogen	A measure of how acidic or basic a solution is. Normal body pH ≈ 7.4 .
Ionisation	—	The process where a molecule gains or loses a hydrogen ion (H⁺) , forming charged or uncharged particles.

Glossary and terms related to ionisation of LA

Term	Full Word / Meaning	Simple Definition
Diffusion	—	Movement of molecules from high to low concentration , allowing the anaesthetic to enter the nerve membrane.
Lipid Soluble	—	Able to dissolve in fat ; necessary for crossing the nerve membrane (RN form).
Water Soluble	—	Able to dissolve in water ; necessary for action inside the nerve (RNH ⁺ form).
Onset Time	—	How quickly the anaesthetic starts working; depends on how much is in RN form (non-ionised).
Infection (Acidic Tissue)	—	Lowers tissue pH → increases RNH ⁺ → less RN → slower or failed anaesthesia .

Dissociation Constants (pKa)

- The dissociation of LA is determined by their pKa and the pH of the tissue.
- The pKa is described by the Henderson-Hasselbalch equation.
- The pKa is a measure of the affinity of a molecule for hydrogen ions (H^+). When the pH of the solution has the same value as the pKa of the local anesthetic, exactly 50% of the drug exists in the RNH^+ form and 50% in the RN form.
- As rate of diffusion across the nerve sheath and nerve membrane is related to the proportion of non-ionised drug,
- Local anaesthetics with low pKa have a fast onset of action, and local anaesthetics with a high pKa have a slow onset of action

Understanding the Balance

- RNH^+ (ionised): water soluble, active inside the nerve but cannot cross the membrane.
- RN (non-ionised): lipid soluble, crosses membrane easily.
- Tissue pH and drug pKa determine which form dominates.

Understanding the Balance

- learning points;

👉 Normal tissue (pH ~ 7.4): more RN $\rightarrow$ fast onset.

👉 Infected tissue (acidic): more RNH⁺ $\rightarrow$ weak or failed anaesthesia.

👉 Drugs with pKa near 7.4 (e.g., lidocaine) act faster than higher pKa drugs (e.g., bupivacaine).

Dissociation Constants (pKa)

Term	Meaning	Property	Function
RNH⁺	Ionised form	Water soluble	Blocks nerve inside cell
RN	Non-ionised form	Lipid soluble	Enters nerve cell
pKa	pH where 50% is RNH ⁺ & 50% is RN	Fixed for each drug	Determines onset speed

Lipid solubility

- Lipid solubility and potency are closely related.
- The lipid solubility of local anaesthetics is expressed as the partition coefficient,
- the partition coefficient which is defined as the ratio of concentrations when local anaesthetic is dissolved in a mixture of lipid and aqueous solvents.

Lipid solubility

- Greater lipid solubility enables more rapid diffusion through lipid membranes to reach their site of action, influencing the speed of onset,
- The greater lipid solubility gives a greater volume of distribution.
- The greater lipid solubility produce more effective conduction blockade at lower concentrations (lower percentage solutions or smaller volumes deposited) than is produced by less lipid-soluble local anesthetics.

Vasoactivity

- Affects both the anesthetic potency and the duration of anesthetic drug
- Injection of local anesthetics, such as procaine, with greater vasodilating properties increases perfusion of the local site with blood.
- The injected local anesthetic is absorbed into the cardiovascular circulation more rapidly and is carried away from the injection site and from the nerve,
- thus, providing for a shortened duration of anesthesia, as well as decreased potency of the drug.

Protein binding

- Local anaesthetics with high protein binding to α_1 -acid glycoprotein have a longer duration of action and lower bioavailability.
- Hypoxia, hypercarbia, and acidaemia all decrease protein binding, and increase the risk of toxicity.
- Children younger than 6 months have less protein binding capacity.

Protein binding

- The degree of protein binding of the local anesthetic molecule is responsible for the duration of anesthetic activity,
- Proteins constitute approximately 10% of the nerve membrane, and
- Some local anesthetics (e.g., etidocaine, ropivacaine, bupivacaine) possessing a greater degree of protein binding and possess a longer duration of clinical activity.

Factors Affecting Local Anesthetic Action

Factor	Action Affected	Description
pK _a	Onset	Lower pK _a = More rapid onset of action, more RN molecules present to diffuse through nerve sheath; thus onset time is decreased
Lipid solubility	Anesthetic potency	Increased lipid solubility = Increased potency (e.g., procaine = 1; etidocaine = 140) Etidocaine produces conduction blockade at very low concentrations, whereas procaine poorly suppresses nerve conduction, even at higher concentrations
Protein binding	Duration	Increased protein binding allows anesthetic cations (RNH ⁺) to be more firmly attached to proteins located at receptor sites; thus duration of action is increased
Nonnervous tissue diffusibility	Onset	Increased diffusibility = Decreased time of onset
Vasodilator activity	Anesthetic potency and duration	Greater vasodilator activity = Increased blood flow to region = Rapid removal of anesthetic molecules from injection site; thus anesthetic potency and duration are decreased
From Cohen S, Burns RC: Pathways of the pulp, ed 6, St Louis, 1994, Mosby.		

Pharmacodynamic properties

- Like other drugs, LA, but regardless of the route through which they are administered, must ultimately enter the systemic circulation in a sufficient high concentrations
- LA must attain therapeutic blood levels in their target organ before they can begin to exert a clinical action.
- LA, cease to provide a clinical effect when they are absorbed from the site of administration into the circulation.
- LA redistribution considered prime factor in their termination of action of when used for pain control from their target organs

Pharmacology of LA Absorption

- The absorption is dependent on the site of injection, rate of injection, dosage and vasoactivity of the injectate
- A two significant clinical effect of Vasodilation property worth studying;
 - An increase in the rate of absorption of the local anesthetic drug into the blood,
 - thus decreasing the duration and depth of pain control,
 - while increasing the anesthetic blood (or plasma) concentration and its potential for overdose (toxic reaction).
- In order to localize the site of action and decrease absorption in the blood stream following injection, vasoconstrictor agents are used

Pharmacology of LA Distribution

- Tissue distribution tends to be proportional to the tissue/blood partition coefficient of the local anaesthetic, and the mass and perfusion of the tissue
- Most agents once absorbed into the circulation are readily distributed from blood to well perfused tissues (e.g. brain, heart, liver, etc.)
- As time progresses, local anesthetic partitions into less well perfused tissues[i.e. muscle, fat] as well as into clearance organs[i.e., liver and kidney]
- Can cross placenta and occasionally may depress heart of fetus.

Partition Coefficient of Local Anaesthetic

Definition:

The **partition coefficient** is the ratio of a drug's solubility in **lipid** (oil) compared to its solubility in **water**.

$$\text{Partition Coefficient} = \frac{\text{Concentration of drug in lipid phase}}{\text{Concentration of drug in aqueous phase}}$$

- **Significance in Local Anaesthesia:**

- Local anaesthetics must **cross the nerve membrane**, which is **lipid-rich**.
- Therefore, **higher lipid solubility** (higher partition coefficient) → **better membrane penetration** → **faster onset** and **greater potency**.
- However, very high lipid solubility can also → **increased toxicity** (drug stays longer in tissues).

Pharmacology of LA Distribution

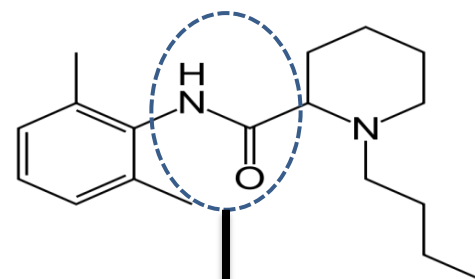
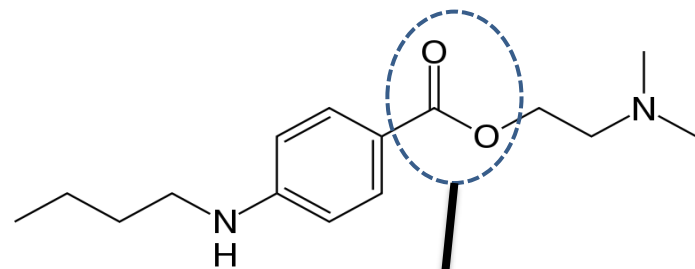
- The plasma concentration of a local anesthetic has a significant bearing on the potential toxicity of the drug.
- Distribution of the local anesthetic is influenced by the following factors:
 - Rate at which the drug is absorbed into the CVS
 - Rate of distribution of the drug from the circulation to the tissues
 - Elimination of the drug through metabolic or excretory process

LA Biotransformation, and Detoxification

- Esters inactivated by hydrolysis in blood and/or liver by pseudocholinesterase; products may be conjugated and/or excreted.
- Individuals with genetically based defects in pseudocholinesterase are unusually sensitive to esters.
- Amides metabolized in liver by oxidation (e.g. N-demethylation), and hydrolysis by liver amidase; products conjugated and excreted.
- Severe hepatic disease may reduce metabolism and lead to toxic reactions.

LA Biotransformation, and Detoxification

- Both esters and amides are excreted in the urine mainly as metabolites with only a few percent as the unchanged agent.
- Amides usually are present in the urine in a greater percentage than the esters due to complex process of biotransformation.



local
anesthetics

Esters

long action
(tetracaine)

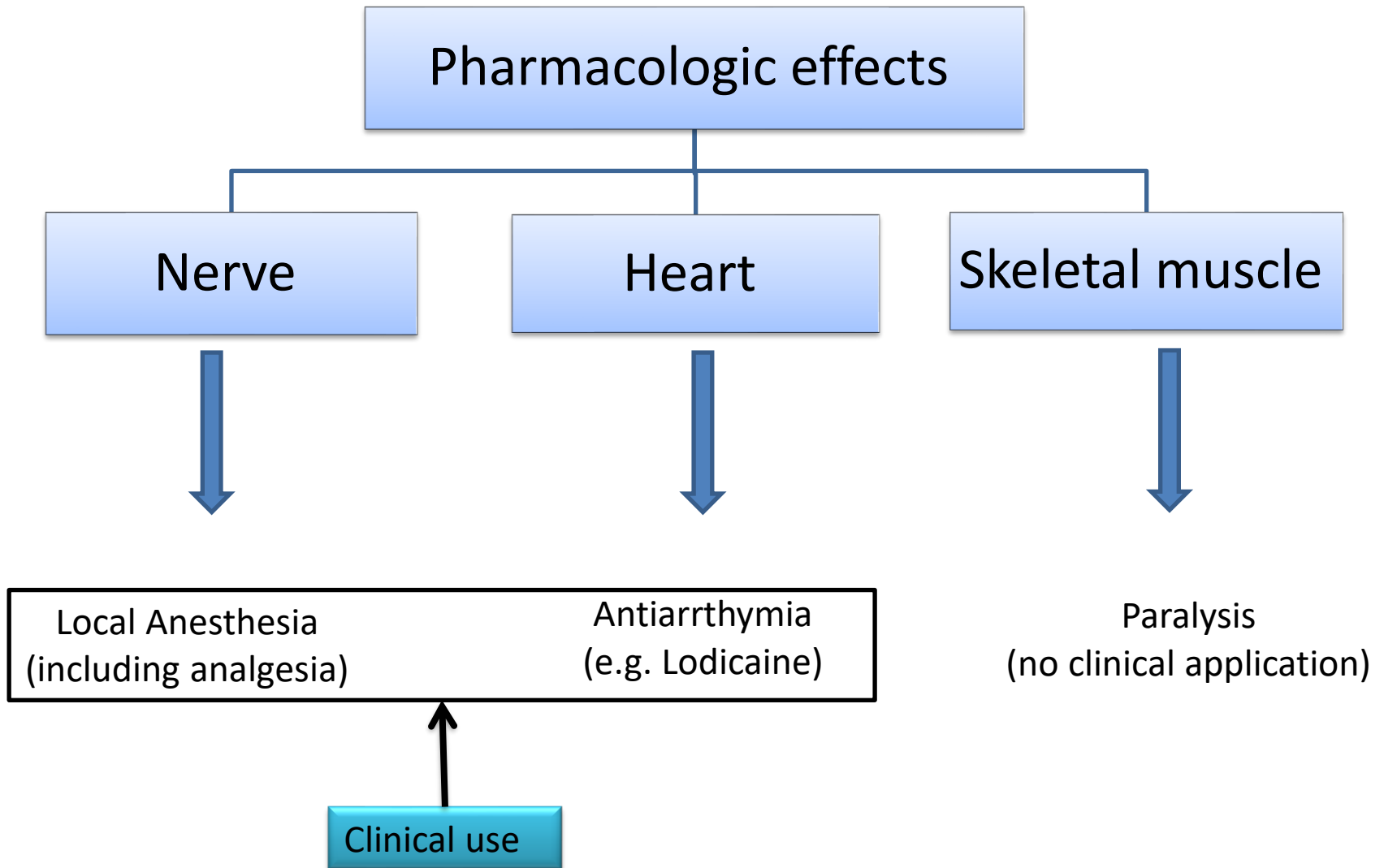
short action
(procaine)

surface action
(benzocaine,
cocaine)

Amides

long action
(bupivacaine,
ropivacaine)

medium action
(lidocaine)



Systemic Actions of Local Anesthetics

- CNS and CVS are greatly susceptible to LA drugs actions.
- Most of the systemic actions of LA are related to their blood or plasma level in the target organ (CNS, CVS).
- The higher the level, the greater will be the clinical actions.

Systemic Actions of Local Anesthetics

- **CNS:** The action of LAs on the CNS is depression. At an overdose levels, the primary clinical manifestation is a generalized tonic-clonic convulsive episode.
 - **Signs:** slurred speech, shivering, muscular twitching.
 - **Symptoms:** numbness of tongue, dizziness, warm skin, light-headedness, visual disturbance (inability to focus), tinnitus, drowsiness, disorientation.
- **CVS:** LA drugs produce a depression of the myocardium. Decrease electrical excitability of the myocardium, decrease the conduction rate, and decrease the force of contraction.

Systemic Actions of Local Anesthetics

- **Respiratory:**
 - At non-overdose levels they have a direct relaxant action on bronchial smooth muscle.
 - At overdose levels they may produce respiratory arrest as a result of generalized CNS depression.

Systemic Actions of Local Anesthetics

- **Peripheral blood vessels:** Cocaine is the only LA that produces vasoconstriction.
 - All other LAs produce vasodilation. This results in an increased blood flow to and from the site of LA deposition.
 - The increase in local blood flow increases the rate of drug absorption. This results in:
 1. Decreased duration of LA action.
 2. Increased bleeding in the treatment area.
 3. Increased anesthetic blood levels (overdose).
- **BP:** LA causes hypotension.

Method of administration of LA

Method of adm.	Definition	e.g. Clinical use
Surface anesthesia	application LA to the surface of the skin or mucosa	Eye surgery Dentistry, Surgery of skin
Infiltration anesthesia	injection of LA into the tissue	minor surgical and dental procedures
Nerve block	Injected of LA in the vicinity of major nerve or major branch nerve	surgical, dental, and diagnostic procedures and for pain management
epidural anesthesia	injected of LA into the epidural space where it acts primarily on the spinal nerve roots	Labor pain Postoperative pain
Spinal anesthesia	injected LA into the cerebrospinal fluid, usually at the lower back, where it acts on spinal nerve roots and part of the spinal cord.	operations below the umbilicus and Leg
Sympathetic block	injected LA around sympathetic nerves	Block some kind of pain (Cancer)

Desirable Properties of Local Anesthetics

- It should not be irritating to the tissue to which it is applied.
- It should not cause any permanent alteration of nerve structure.
- Its systemic toxicity should be low.
- It must be effective regardless of whether it is injected into the tissue or is applied locally to mucous membranes.
- The time of onset of anesthesia should be as short as possible.
- The duration of action must be long enough to permit completion of the procedure yet not so long as to require an extended recovery.
- It should have potency sufficient to give complete anesthesia without the use of harmful concentrated solutions.
- It should be relatively free from producing allergic reactions.
- It should be stable in solution and should readily undergo biotransformation in the body.
- It should be sterile or capable of being sterilized by heat without deterioration